

1. Project Vision & Architecture Mapping

This project modernizes a traditional legacy on-premises Windows Server deployment into a highly resilient, cloud-native hybrid ecosystem. By moving workloads to a structured Cloud Virtual Network (VPC/VNet) and incorporating an enterprise Linux distribution (e.g., Ubuntu Server or Rocky Linux), students gain critical skills in multi-OS integration, modern identity management, reverse-proxy load balancing, containerized high availability, and automated data governance.

TRADITIONAL WINDOWS REQUIREMENT	CLOUD-NATIVE & LINUX HYBRID TRANSFORMATION
Domain Environment: On-Premises PDC (192.168.1.2) & ADC (192.168.1.3), Domain FINAL.LOCAL	Hybrid Identity Infrastructure: Distributed Windows instances deployed within private Cloud Subnets. Integration of an enterprise Linux Host (LINUX-SRV) joined securely to the active directory realm via <code>realmd</code> and <code>SSSD</code> .
Networking & DHCP: Local DHCP Scopes, exclusions, and failover roles.	Cloud Fabric SDN: Network managed natively via Cloud Custom Router/ VPC DHCP Option Sets. Broadcast scopes are transformed into managed private subnets with strict Cloud Network Security Groups (NSGs) blocking the matching IP ranges (e.g., 192.168.1.80/32 to 192.168.1.85/32).
Web Server & Load Balancing: Two IIS instances with local DNS Round Robin.	Layer 7 Reverse Proxy: LINUX-SRV acts as the central ingress point executing an Nginx Reverse Proxy , load balancing incoming application requests across the downstream IIS web engines with health-checks.
File Shares, Quotas & Screening: Windows File Server Resource Manager (FSRM) on native NTFS paths.	AD-Integrated Samba Share: Hosted on LINUX-SRV using <code>smbd</code> natively mapping Active Directory ACLs. Employs Linux advanced file screening rules (<code>veto files</code>) to block forbidden binaries/media, and group quotas via Linux disk tools.
Hyper-V & Hyper-V Replica: Nested virtualization and VM replication across hosts.	Cloud Containerization & Resilience: Replacement of nested virtualization with a Docker Engine environment hosted on Linux. Application consistency and data persistence are maintained using automatic configuration sync and block-level replication mechanisms.
Server Backup: Local Windows Server Backup pointing to a secondary network volume.	Automated Backup Pipeline: Shell scripts orchestrated on LINUX-SRV using cron jobs to capture system states, stream encrypted archives every night at 23:00, and push them directly onto a secure, remote object storage bucket or isolated archival server.

2. Network Topology & Base Subnetting Matrix

The physical layout transitions to a secure Software-Defined Network (SDN). All infrastructure lives within an isolated cloud network block. Broadcast constraints are native to the cloud fabric, requiring manual resource mapping for static reservations.

- **Network Address Space:** 192.168.1.0/24
- **Cloud Default Gateway:** 192.168.1.1
- **Primary Active Directory Domain Controller (PDC):** 192.168.1.2
- **Additional Domain Controller Core (ADC):** 192.168.1.3
- **Enterprise Linux Integration Node (LINUX-SRV):** 192.168.1.4
- **Automated Cloud Dynamic Assignment Range:** 192.168.1.40 to 192.168.1.230
- **Strict Exclusion / Firewall Block Zone:** 192.168.1.80 to 192.168.1.85 (Simulate by deploying Cloud Firewalls / Network ACLs to reject all ingress/egress traffic on these addresses).

3. Phase-by-Phase Technical Specifications

PHASE 1: HYBRID DOMAIN ESTABLISHMENT & LINUX INTEGRATION

1. **Windows Infrastructure:** Build a fresh Active Directory Domain Services forest with the root FQDN set to FINAL.LOCAL on the PDC host. Configure an Additional Domain Controller on a minimal Windows Server Core installation (ADC), and join it to the existing domain topology.
2. **Organizational Units (OUs) & Accounts:** Programmatically create separate OUs for each core business unit: HR, Sales, Dev, and IT. Generate respective security groups and corresponding test accounts within each partition.
3. **Linux Realm Admittance:** Configure the LINUX-SRV network configuration to utilize 192.168.1.2 and 192.168.1.3 as its primary DNS resolvers. Install the required integration subsystems:

```
sudo apt-get update && sudo apt-get install -y sssd sssd-tools realmd krb5-user  
samba-common-bin packagekit
```

4. **Domain Join Verification:** Request a Kerberos ticket for the domain administrator account and join the FINAL.LOCAL active directory framework:

```
sudo kinit administrator  
sudo realm join --user=administrator FINAL.LOCAL
```

Configure /etc/sss/sss.conf to permit fully qualified domain logins and ensure home directories are generated on the first authentication cycle via pam_mkhomedir.so.

PHASE 2: ENTERPRISE SECURITY POLICY ALIGNMENT

Security baselines must apply globally. Account governance must be controlled dynamically from a single source.

- **Password Constraints:** Define an Active Directory Group Policy Object (GPO) or Fine-Grained Password Policy specifying a maximum life parameter of $T_{max} = 60$ days, a minimum physical structure length of $L_{min} = 6$ characters, mandatory execution of complexity requirements, and a history validation threshold storing the last $H = 3$ configurations.
- **Account Lockout Thresholds:** Implement an aggressive security posture forcing a constraint window where accounts are systematically locked for exactly 1 hour following $N = 4$ sequential validation failures.
- **Remote Administration & Access Controls:** Enable Remote Desktop Protocol (RDP) globally on all domain machines through a dedicated computer configuration GPO. Additionally, deploy Remote Assistance rules explicitly identifying members of the IT security group as authorized global helpers.
- **Endpoint Restrictions:** Deploy endpoint hardening GPOs targeting workstation components: completely prohibit access to external removable storage mechanisms, mask the system Task Manager, and fully restrict access to the legacy Control Panel environment.
- **Resilience Controls:** Activate the Active Directory Recycle Bin feature via administrative PowerShell modules to ensure object recovery metrics are optimized.

Instructor Note: When checking cross-platform policy enforcement, verify that Linux hosts respect domain account lockouts. SSSD queries the Domain Controller in real-time, executing the active lockout state across SSH and local tty sessions.

PHASE 3: LAYER 7 REVERSE PROXY & WEB INFRASTRUCTURE

Instead of vulnerable DNS round-robin routing which lacks infrastructure availability metrics, this modern implementation sets up a high-performance Linux proxy layer.

1. **IIS Backends:** Install the Internet Information Services (IIS) web role on both PDC and ADC Core instances. Customize each base landing page to explicitly display the local host identifier for testing purposes.
2. **Nginx Deployment:** Install Nginx on the LINUX-SRV platform to manage external application requests.
3. **Reverse Proxy Setup:** Modify the configuration at `/etc/nginx/sites-available/default` to declare an upstream grouping that distributes traffic symmetrically across both Windows web hosts:

```
upstream web_backends {
    server 192.168.1.2:80;
    server 192.168.1.3:80;
}

server {
    listen 80;
    server_name www.final.local;

    location / {
        proxy_pass http://web_backends;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }
}
```

4. **DNS Configuration:** Create a canonical host entry (A Record) within the Active Directory DNS management console mapping `www.final.local` to the IP address of the Linux server (`192.168.1.4`). All client interactions pass through the Nginx proxy layer.

PHASE 4: MULTI-OS STORAGE GOVERNANCE & INTELLIGENT FILE SERVICES

File sharing requirements are enforced on the Linux server using Samba, maintaining full integration with Active Directory user authentication and security identifiers.

- **Samba Integration:** Configure Samba on LINUX-SRV to handle domain permissions. Ensure `/etc/samba/smb.conf` references the proper security mode:

```
[global]
    workgroup = FINAL
    realm = FINAL.LOCAL
    security = ADS
    winbind enum users = Yes
    winbind enum groups = Yes

[department_share]
    path = /srv/samba/dept_share
    read only = no
    guest ok = no
    valid users = @HR@FINAL.LOCAL, @Dev@FINAL.LOCAL
```

- **Advanced Data Governance ACLs:** Set the local Linux directory permissions using POSIX Access Control Lists to allow data addition and modifications while explicitly denying data deletion flags to typical consumers:

```
sudo setfacl -R -m g:"Dev@FINAL.LOCAL":rwx,g:"HR@FINAL.LOCAL":rwx /srv/samba/
dept_share
sudo setfacl -R -d -m g:"Dev@FINAL.LOCAL":rwx,g:"HR@FINAL.LOCAL":rwx /srv/samba/
dept_share
sudo chmod +t /srv/samba/dept_share
```

Note: Applying the sticky bit (+t) guarantees users can modify assets but are restricted from deleting items owned by others.

- **Intelligent File Screening:** Replicate Windows FSRM file blocks by integrating a native Samba veto block expression directly into the share configuration to prevent media and execution uploads:

```
veto files = /*.mp3/*.mp4/*.avi/*.exe/*.msi/
delete veto files = yes
```

- **Public Share and Storage Quota Assignment:** Construct a public share directory accessible to all authenticated domain workers. Apply strict Linux file quotas limiting allocation caps to exactly $Q_{max} = 2$ GB per individual domain user account using the `quotacheck` and `edquota` tools.
- **Automated Drive Provisioning:** Author a GPO target executing a logon script or a Group Policy Preference item mapping the Linux-hosted departmental share as a drive letter on all Windows client systems.

PHASE 5: CONTAINERIZATION, HA, & DISASTER RECOVERY

The physical high-availability model transitions away from legacy Hyper-V local structures into a lightweight, cloud-ready container architecture paired with automated workflows.

1. **Docker Container Engine Setup:** Install Docker Engine on the LINUX-SRV node to serve as the core virtualization layer.
2. **Application Isolation:** Build or pull an isolated internal service application container running inside the engine, mapping its external ports to the host's physical networking interface.
3. **Automated Synchronization Workflow (Replica Emulation):** To replicate the disaster recovery profiles of Hyper-V Replica, configure a background synchronization architecture utilizing `rsync` or automated storage snapshot engines. This process captures container volume data and state vectors on LINUX-SRV and securely copies them to an isolated backup path on the ADC server every hour.
4. **Enterprise Backup Schedule:** Implement a production-grade cron utility block on the LINUX-SRV core infrastructure to automate offline data backups. Create a script at `/etc/cron.d/server_backup`:

```
0 23 * * * root tar -czf /mnt/adc_backup/linux_srv_backup_$(date +%F).tar.gz /srv/
samba/ /var/www/html/
```

This executes a complete, compressed archival sweep of all data shares, configuration files, and application directories daily at precisely 11:00 PM, transmitting the final payload directly onto the secondary backup storage target hosted on the ADC volume.

4. Student Verification Checklist & Deliverables

To successfully pass evaluation, students must compile an infrastructure report confirming the operational health of the following components:

#	VERIFICATION TARGET	EXPECTED PROOF / COMMAND OUTPUT
1	Cross-OS Authentication	Successful output from running <code>id user_hr@final.local</code> on the Linux host showing valid Active Directory SIDs and group assignments.
2	Layer 7 Load Balancing	Refreshing <code>http://www.final.local</code> inside a client web browser alternates between displaying the PDC and ADC hostname markers.
3	Samba Deletion Blocker	Attempting to run a delete command on a file inside the department share owned by another domain user returns an <code>Access Denied</code> error.
4	File Screening Filter	Attempting to drop a file named <code>payload.exe</code> or <code>track.mp3</code> into the share path triggers an immediate file block block-action.
5	Data Quota Integrity	Executing <code>repquota -a</code> on the Linux filesystem displays a hard policy constraint threshold set to 2048MB for domain security identities.
6	Automated Backup Tasks	The presence of valid timestamped backup archives inside the designated target storage path on the ADC host.